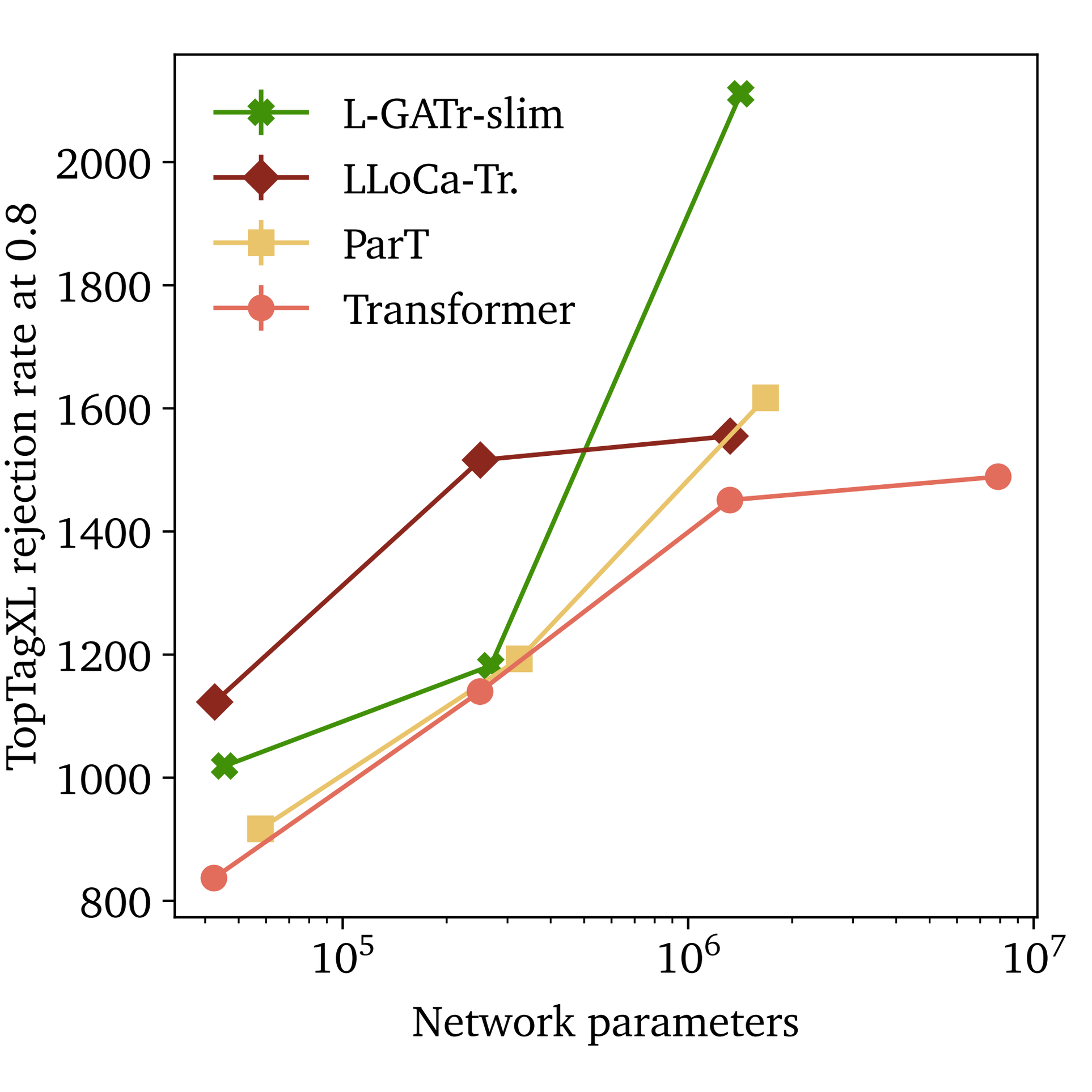
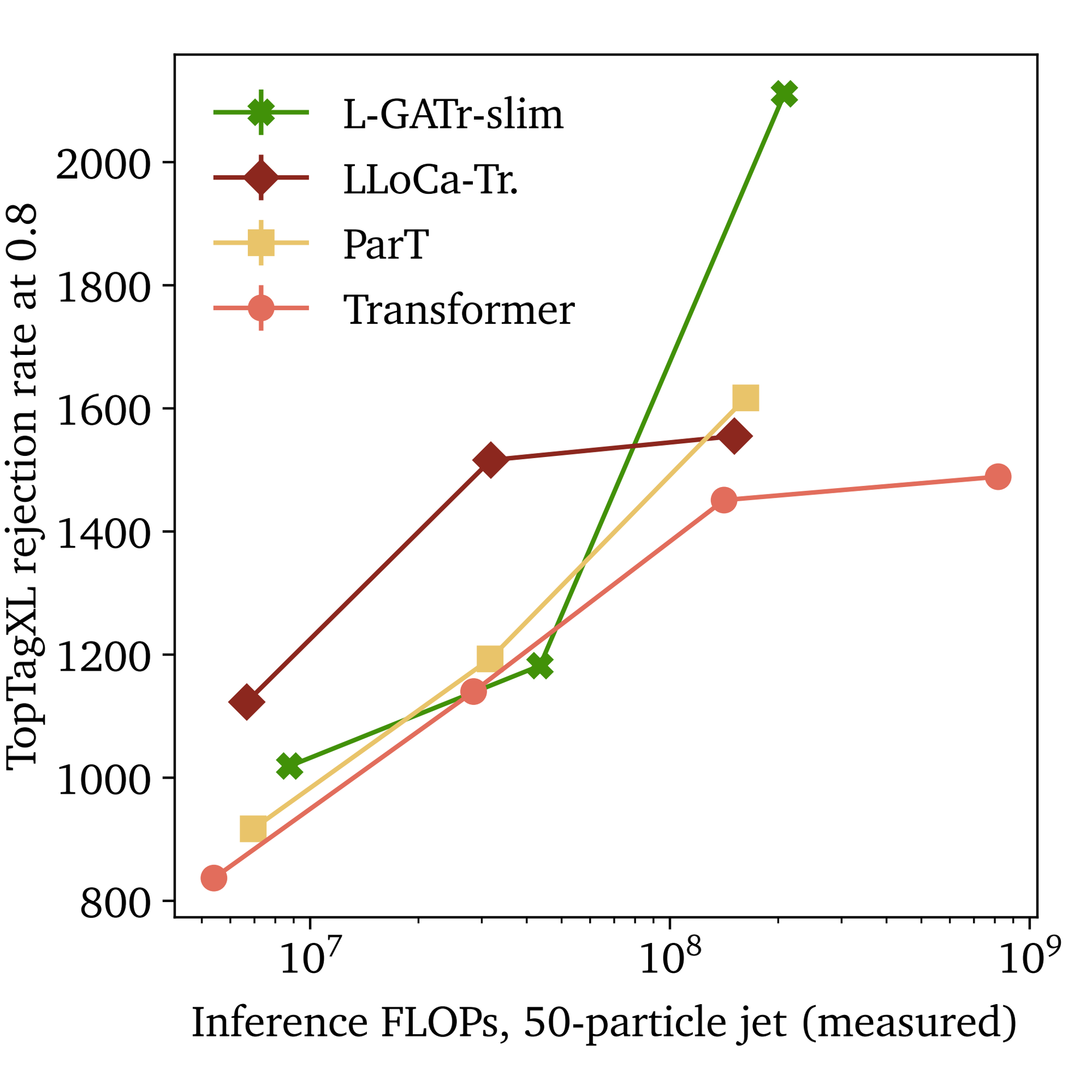
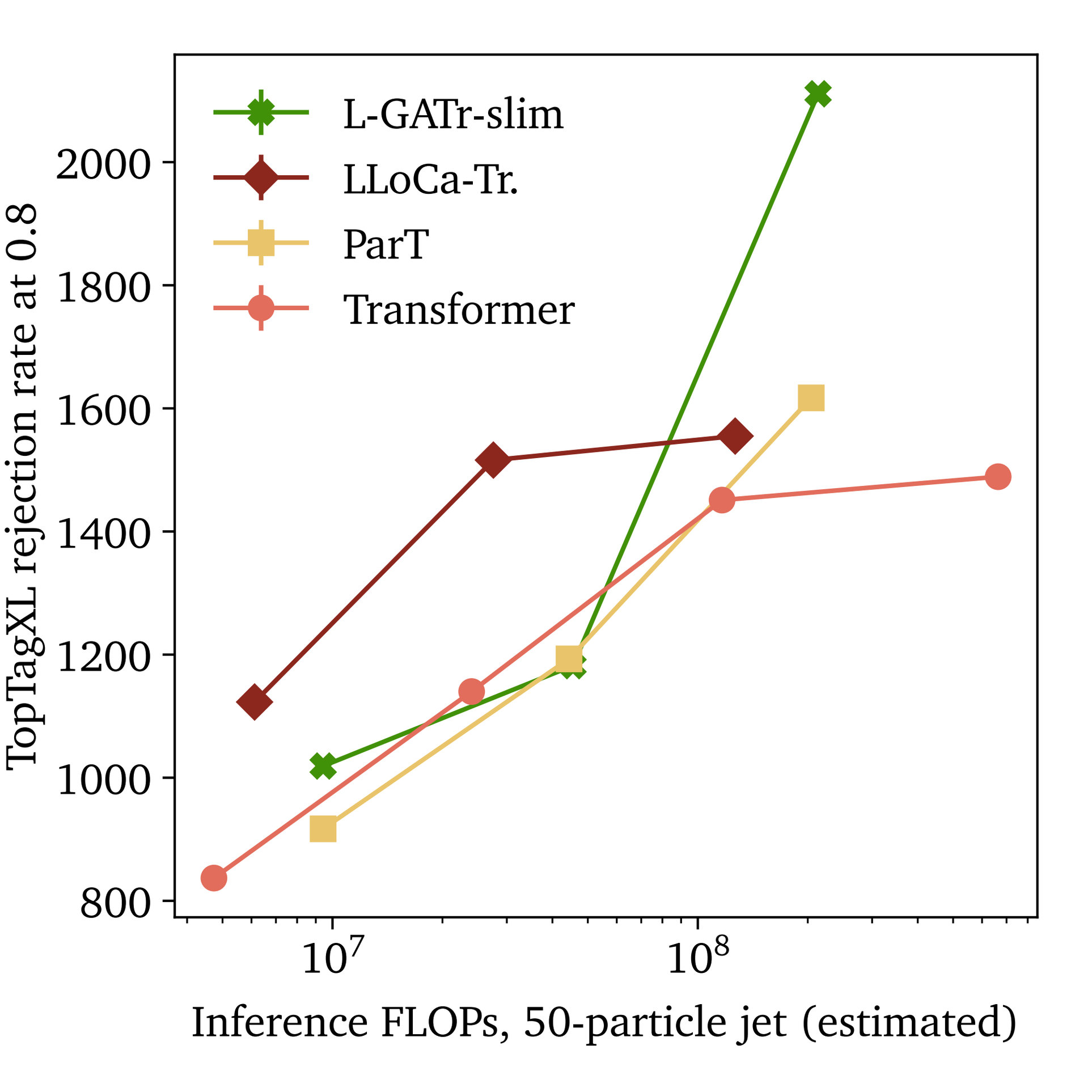
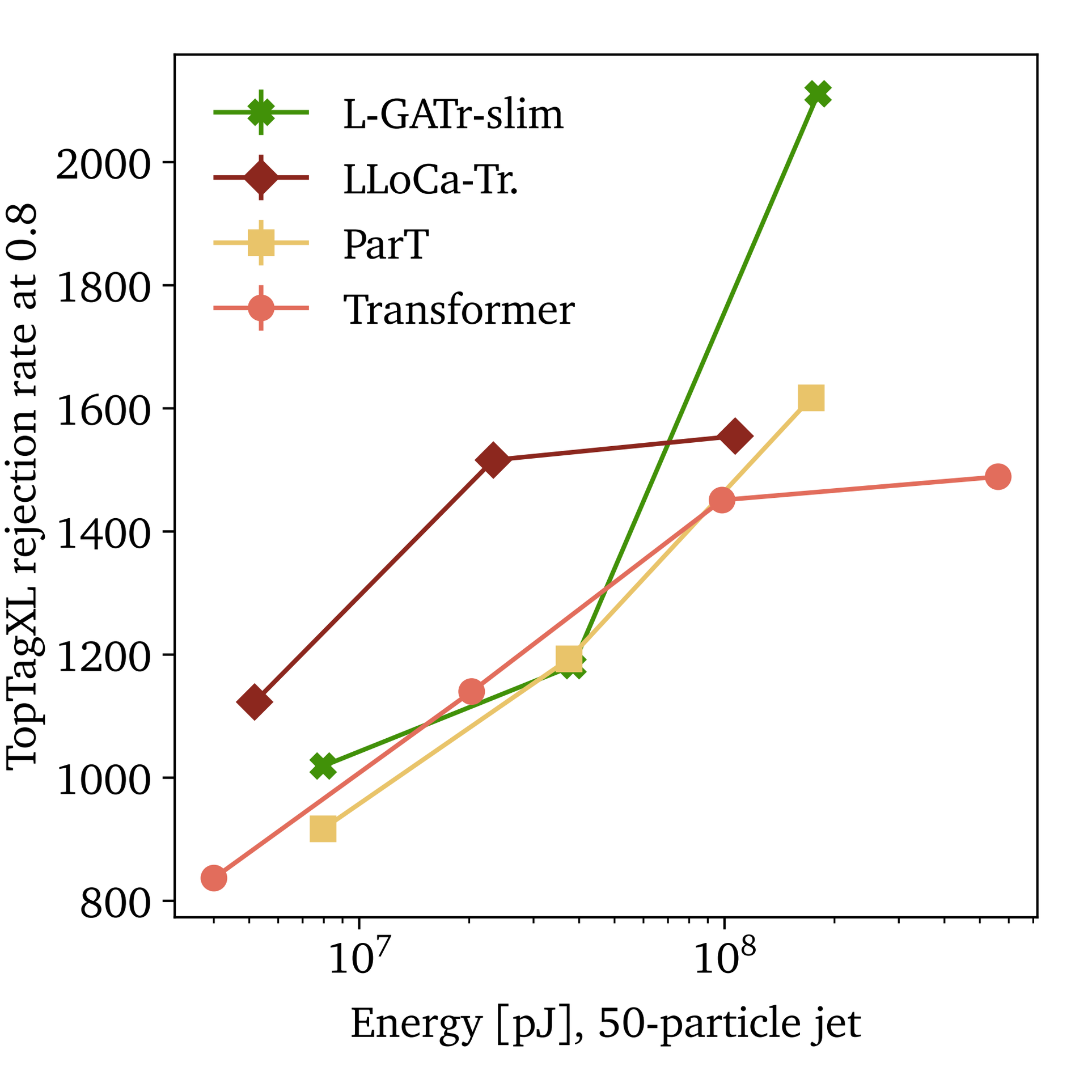
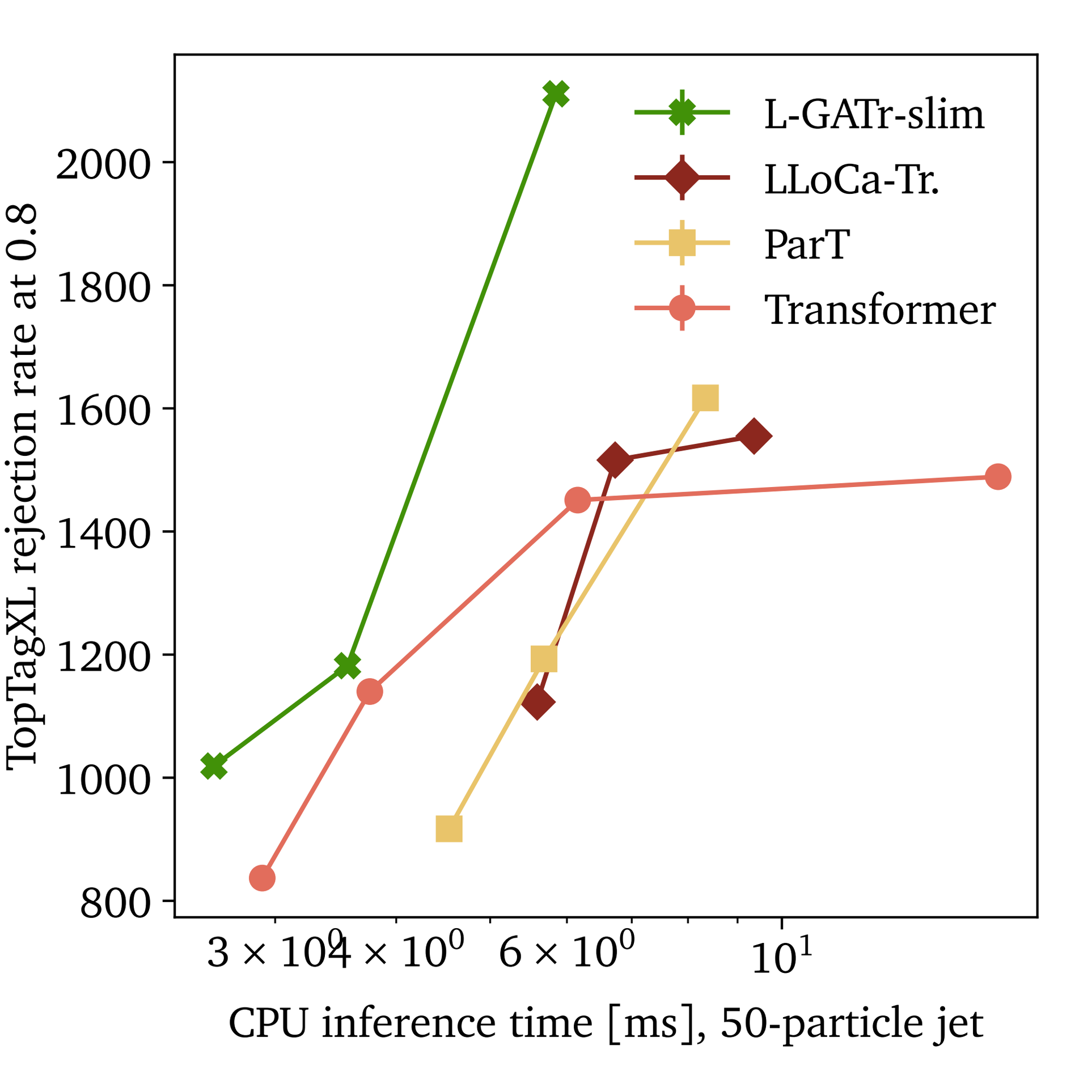
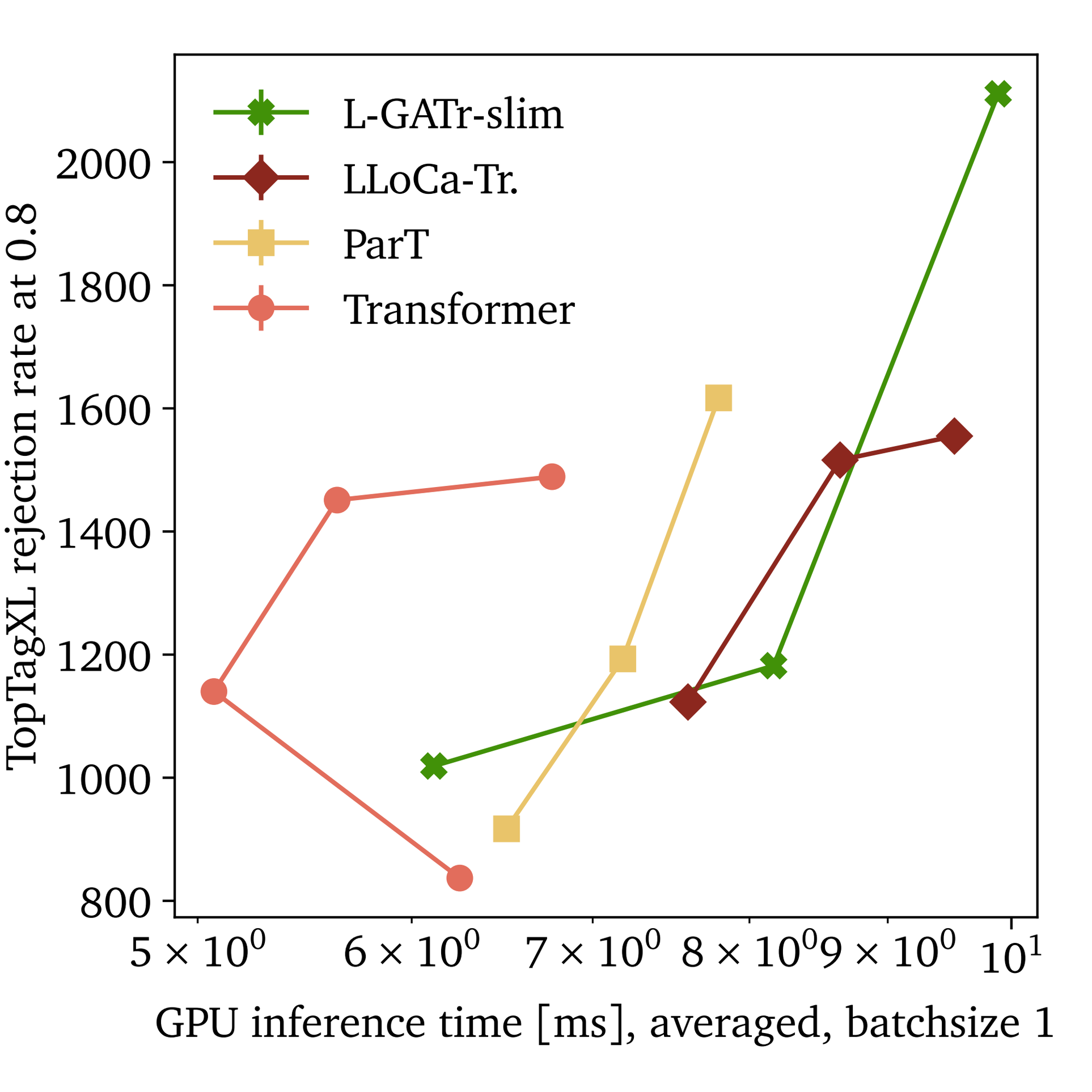

L-GATr-slim
LLoCa-Tr.
ParT
Transformer
2000
1800
1600
1400
1200
1000
800
TopTagXL rejection rate at 0.8
10^5
10^6
10^7
Network parameters

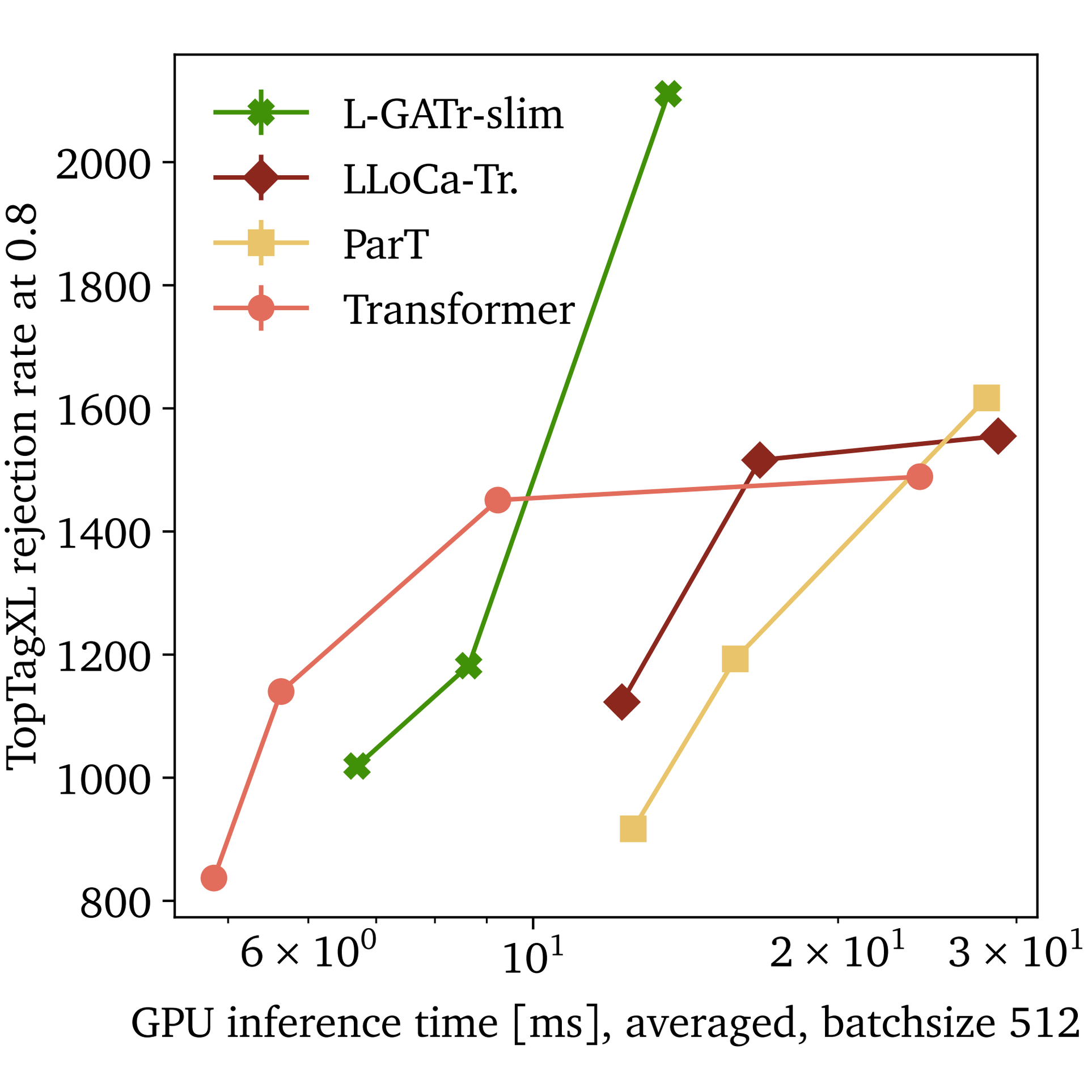

L-GATr-slim
LLoCa-Tr.
ParT
Transformer
2000
1800
1600
1400
1200
1000
800
TopTagXL rejection rate at 0.8
10^7
10^8
10^9
Inference FLOPs, 50-particle jet (measured)

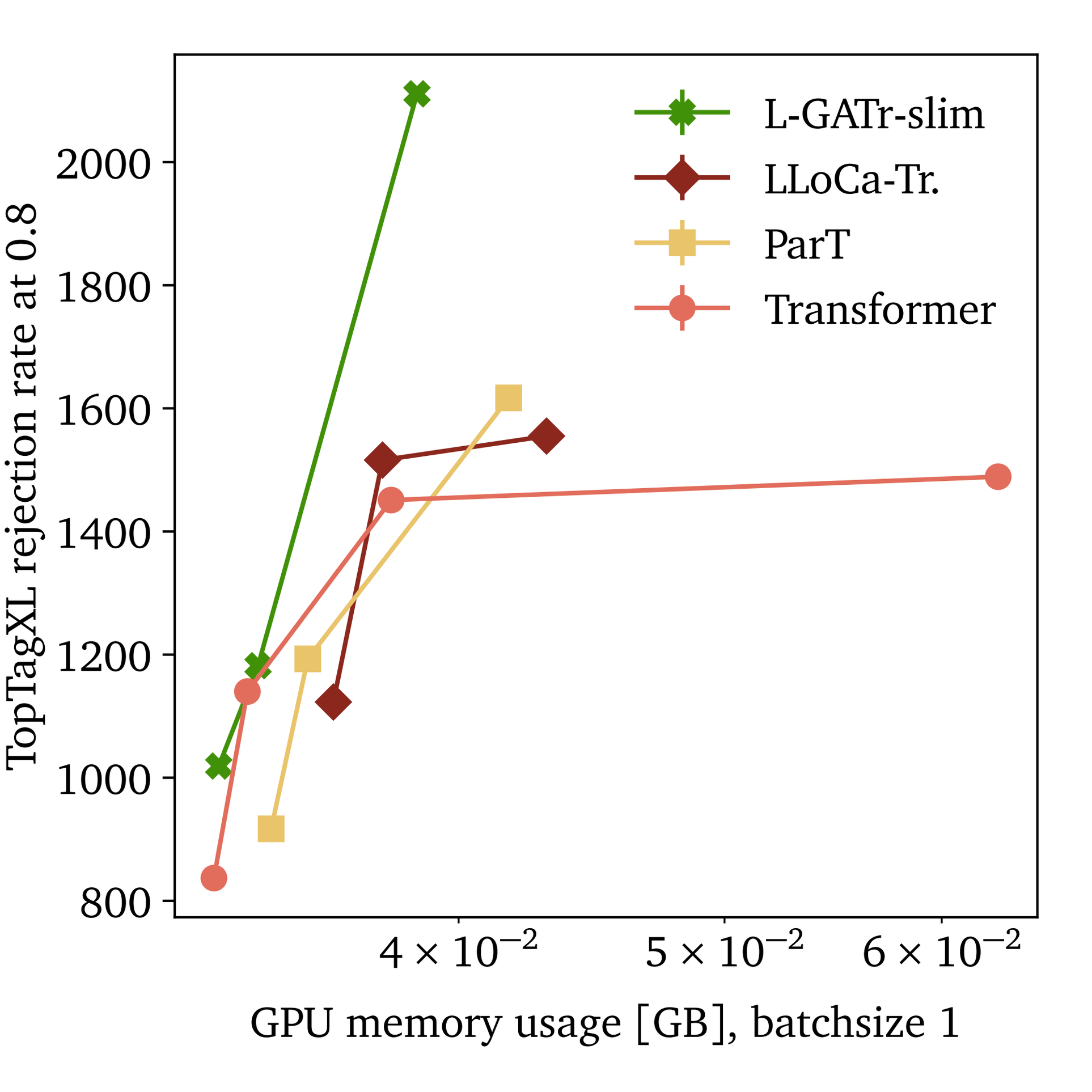

L-GATr-slim
LLoCa-Tr.
ParT
Transformer
2000
1800
1600
1400
1200
1000
800
TopTagXL rejection rate at 0.8
10^7
10^8
Inference FLOPs, 50-particle jet (estimated)

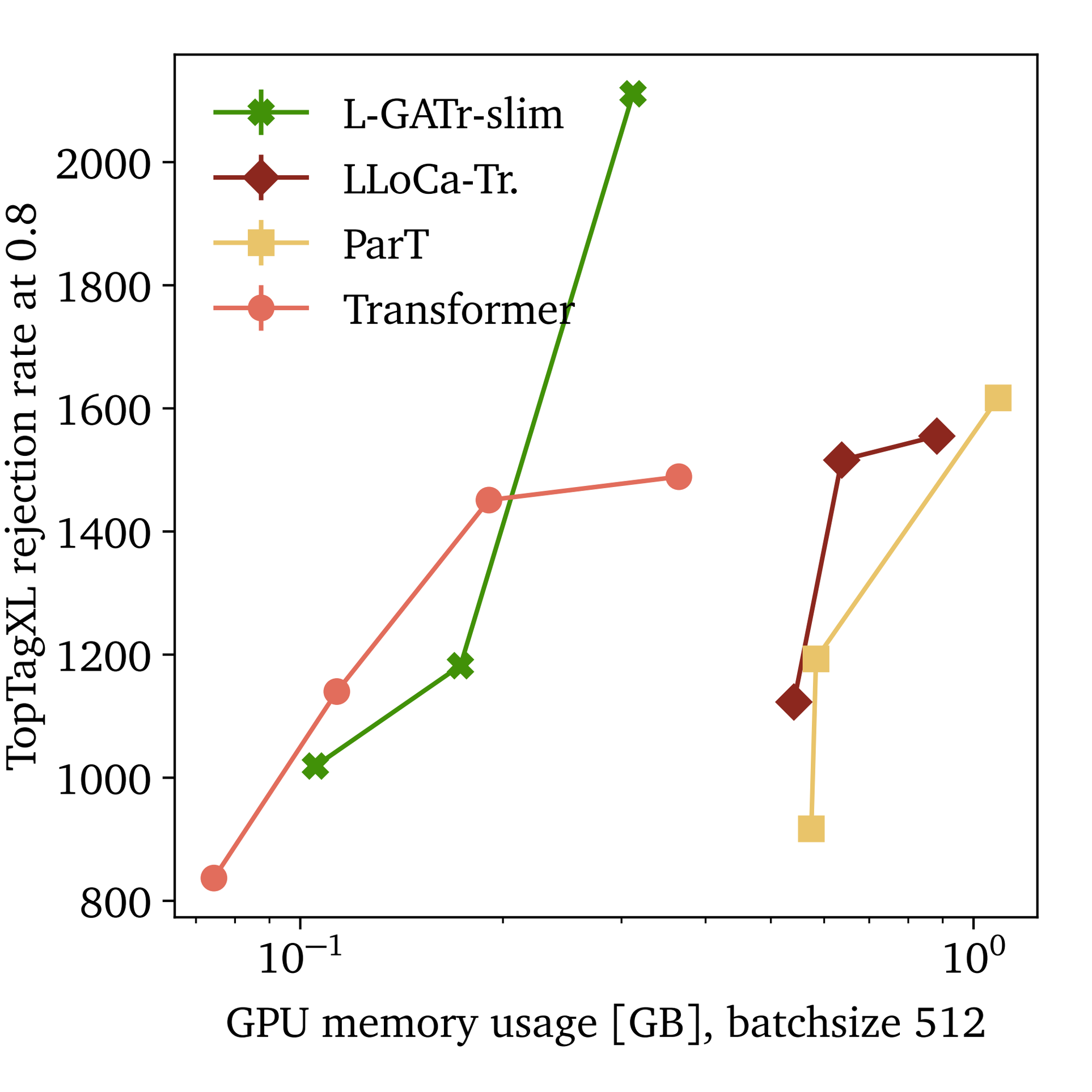

L-GATr-slim
LLoCa-Tr.
ParT
Transformer
2000
1800
1600
1400
1200
1000
800
TopTagXL rejection rate at 0.8
10^7
10^8
Energy [pJ], 50-particle jet

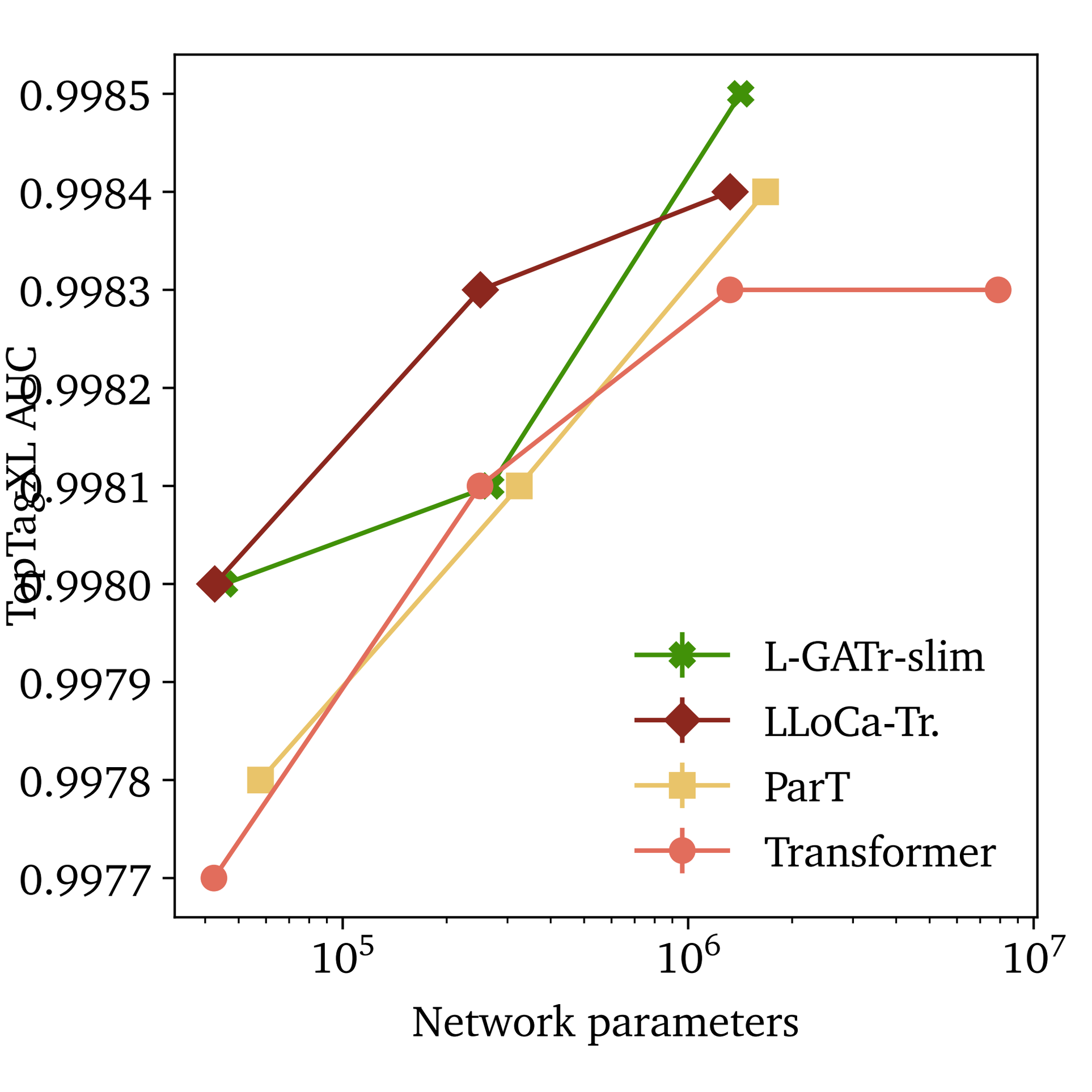

L-GATr-slim
LLoCa-Tr.
ParT
Transformer
2000
1800
1600
1400
1200
1000
800
TopTagXL rejection rate at 0.8
3×10^0
4×10^0
6×10^0
10^1
CPU inference time [ms], 50-particle jet

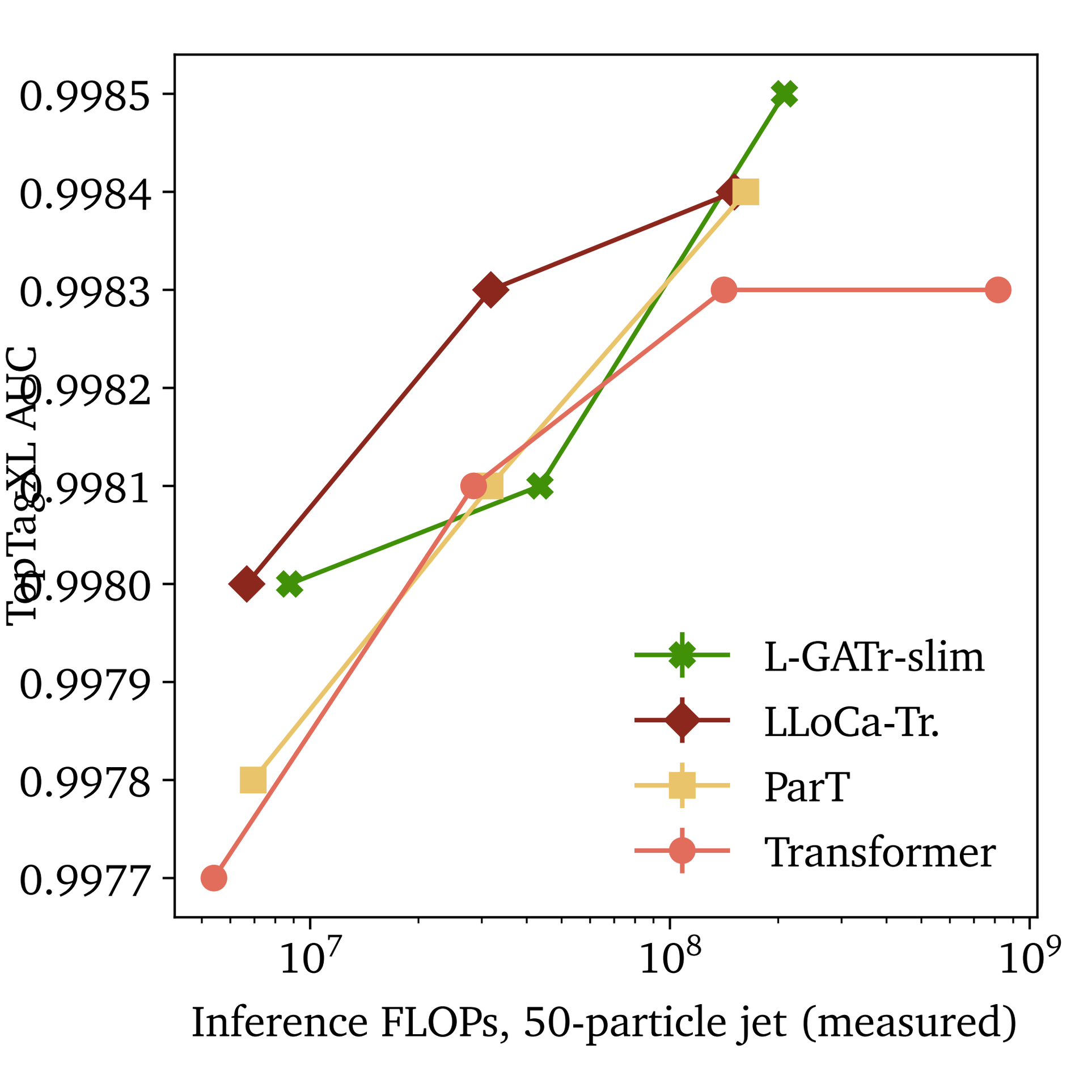

L-GATr-slim
LLoCa-Tr.
ParT
Transformer
2000
1800
1600
1400
1200
1000
800
TopTagXL rejection rate at 0.8
5×10^0
6×10^0
7×10^0
8×10^0
9×10^0
10^1
GPU inference time [ms], averaged, batchsize 1

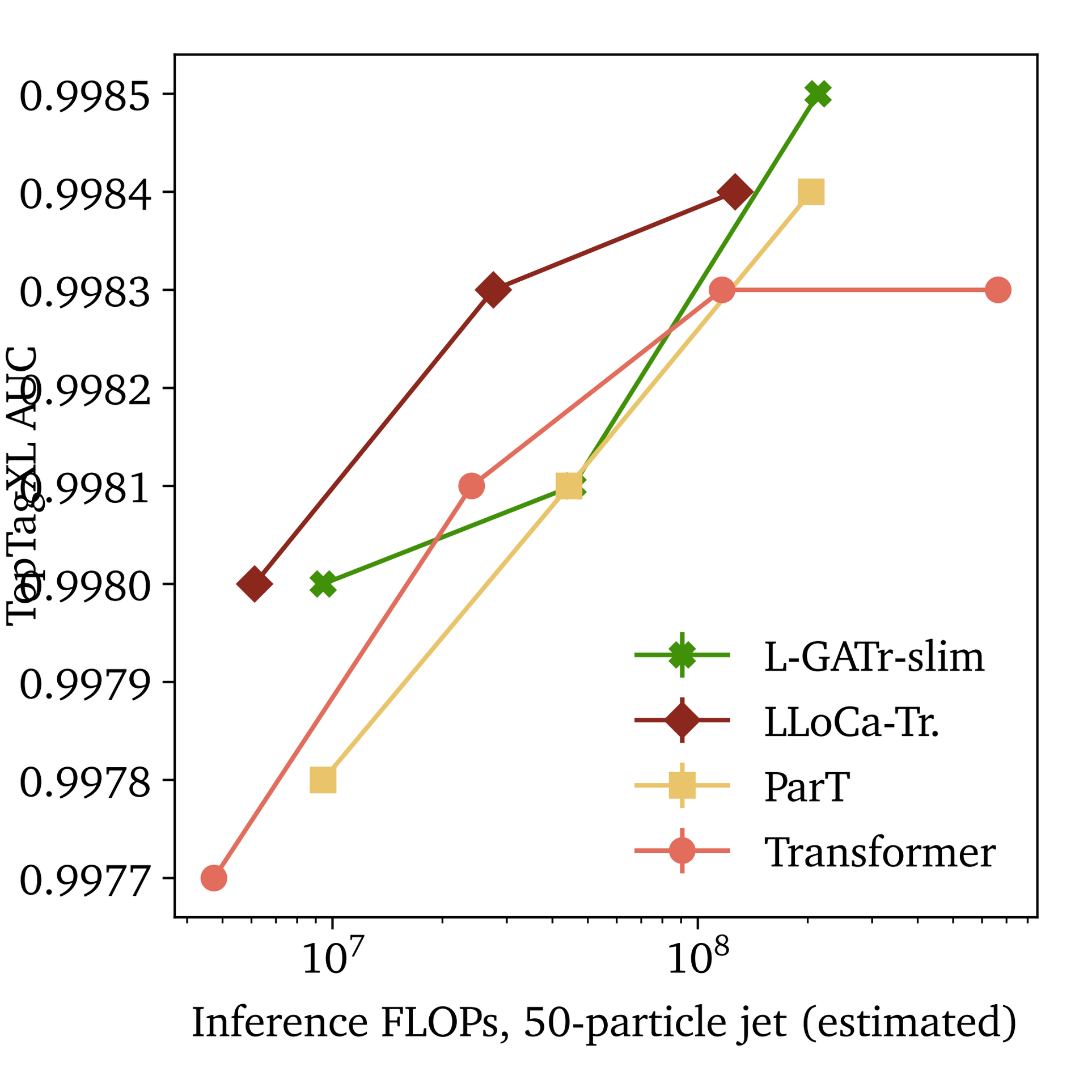

L-GATr-slim
LLoCa-Tr.
ParT
Transformer
2000
1800
1600
1400
1200
1000
800
TopTagXL rejection rate at 0.8
6×10^0
10^1
2×10^1
3×10^1
GPU inference time [ms], averaged, batchsize 512

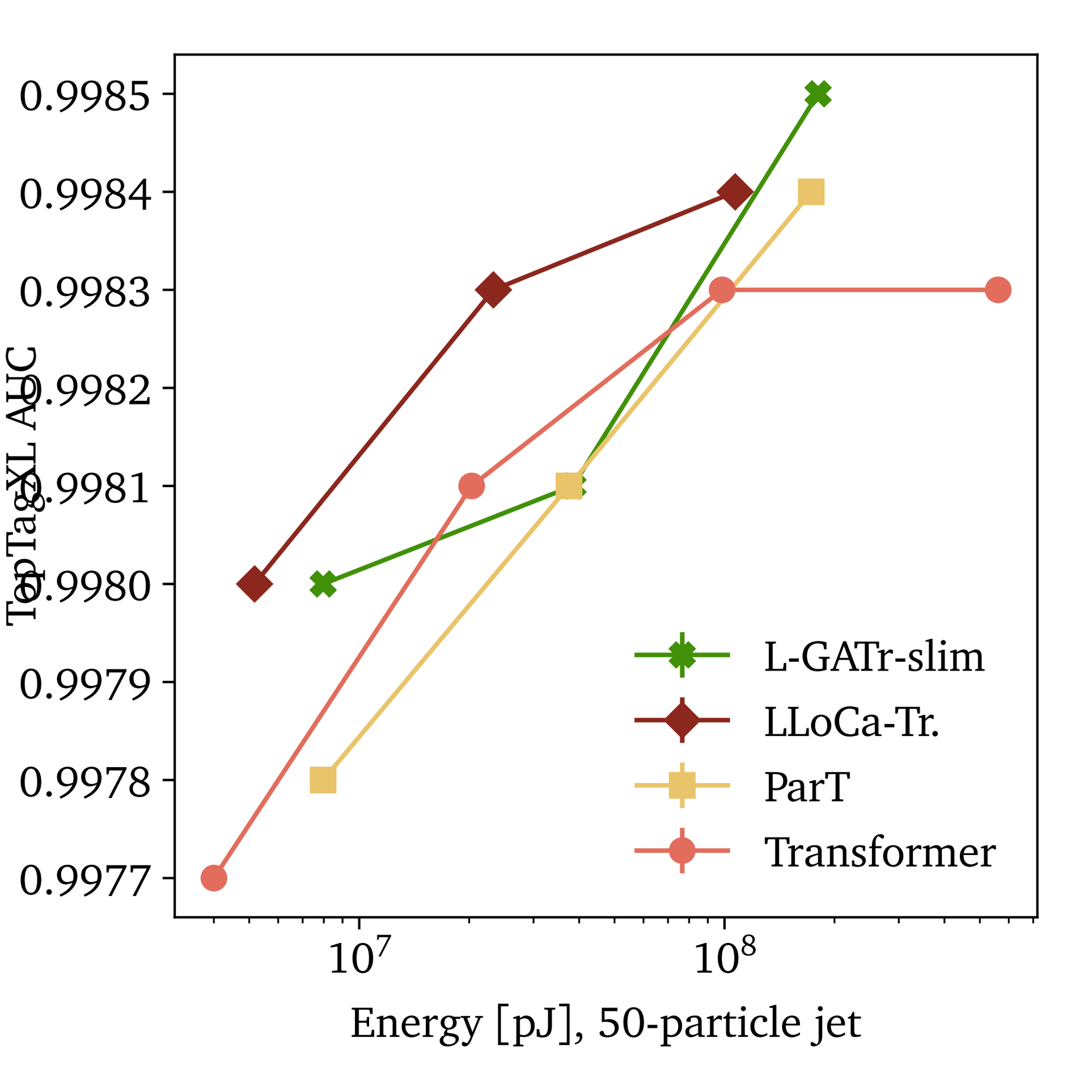

L-GATr-slim
LLoCa-Tr.
ParT
Transformer
TopTagXL rejection rate at 0.8
2000
1800
1600
1400
1200
1000
800
4×10^{-2}
5×10^{-2}
6×10^{-2}
GPU memory usage [GB], batchsize 1

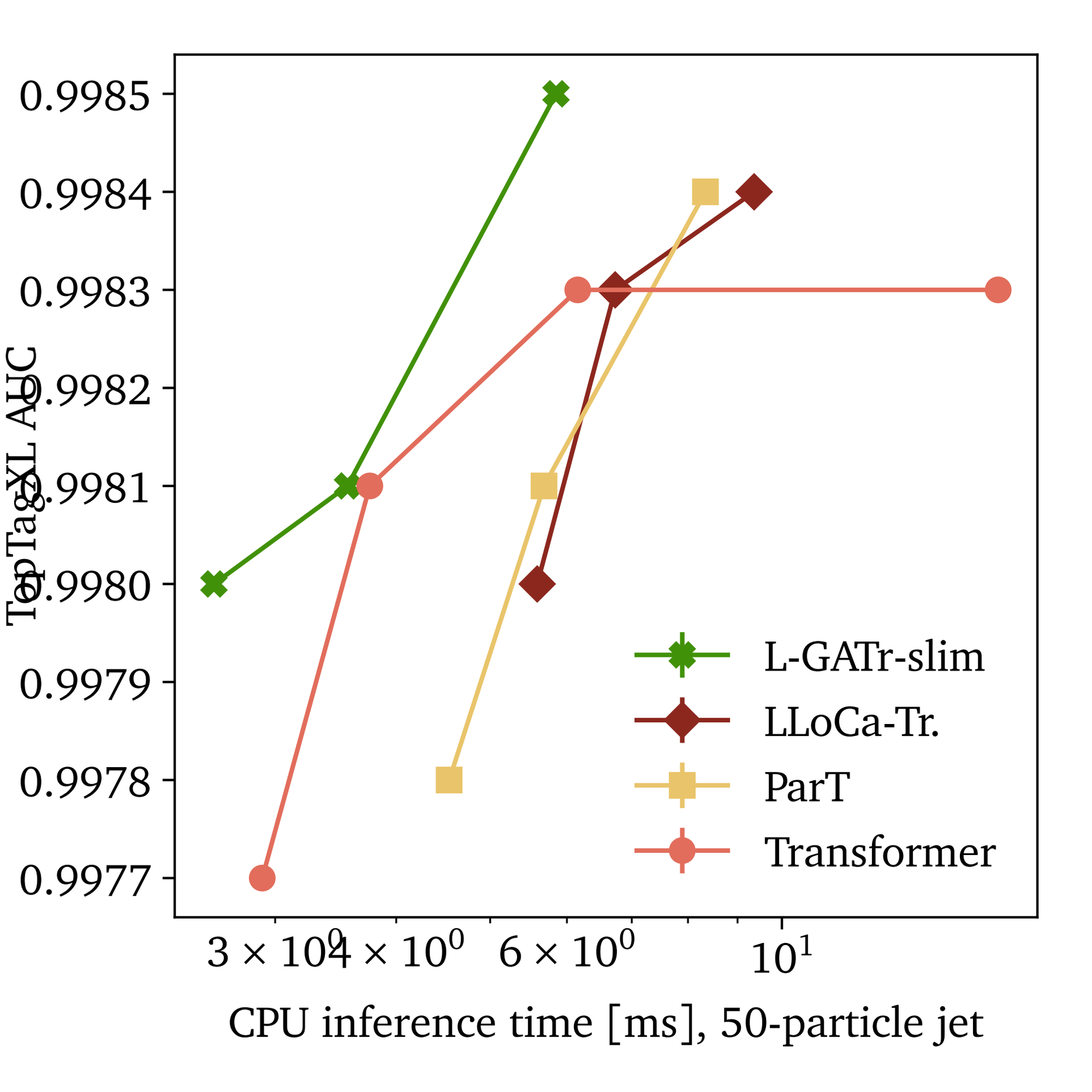

L-GATr-slim
LLoCa-Tr.
ParT
Transformer
2000
1800
1600
1400
1200
1000
800
TopTagXL rejection rate at 0.8
10^{-1}
10^{0}
GPU memory usage [GB], batchsize 512

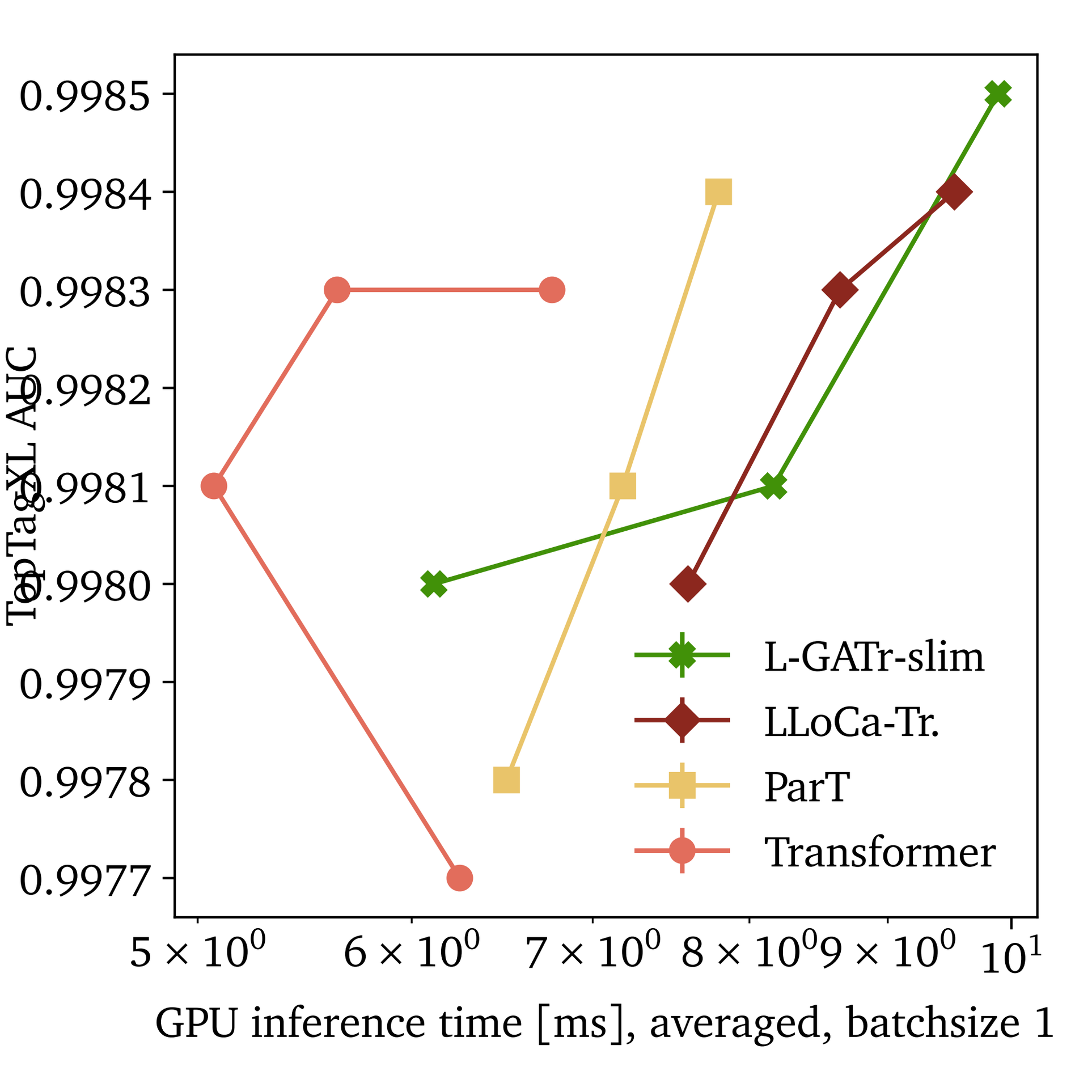

0.9985
0.9984
0.9983
0.9982
0.9981
0.9980
0.9979
0.9978
0.9977
TopTagXL AUC
10^5
10^6
10^7
Network parameters
L-GATr-slim
LLoCa-Tr.
ParT
Transformer

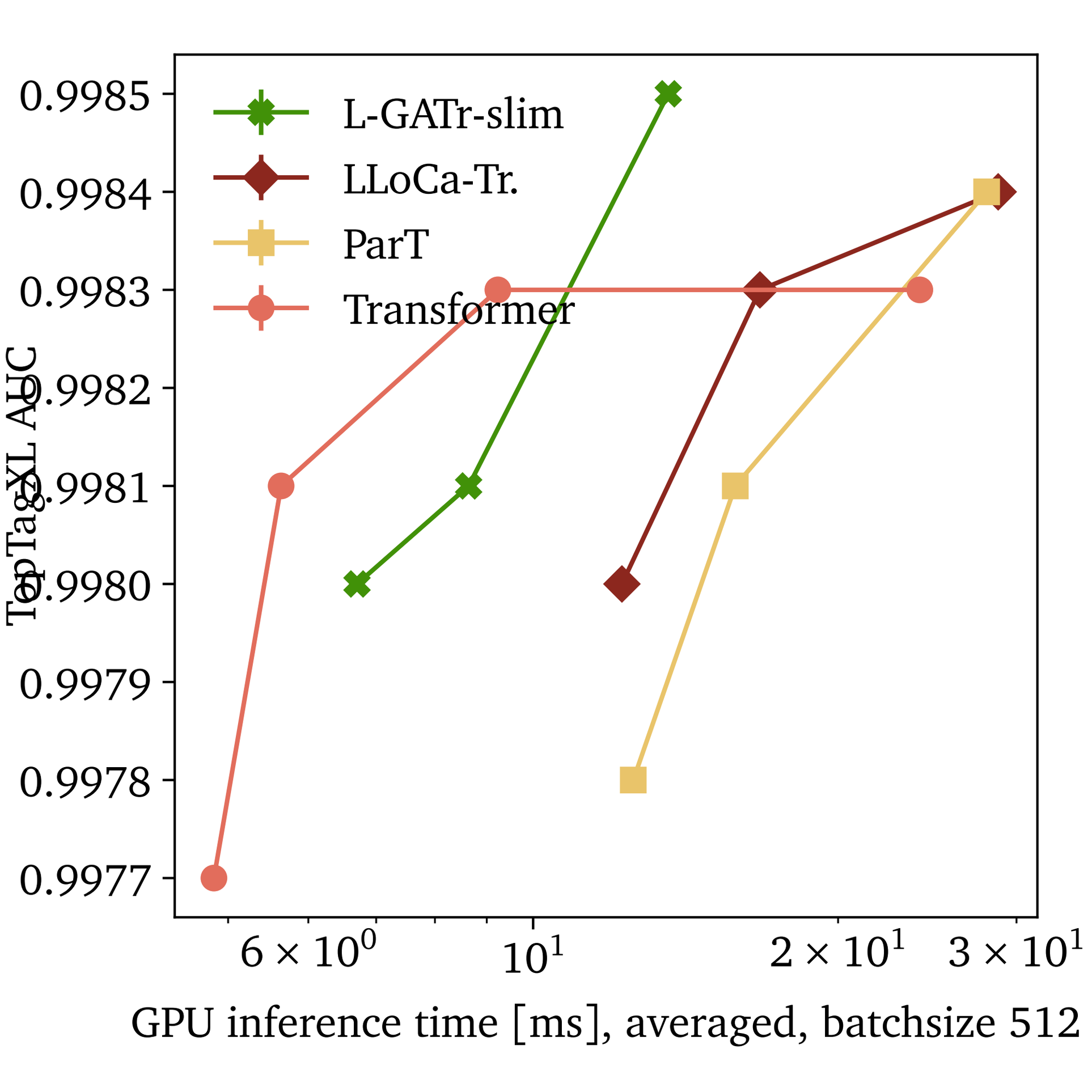

0.9985
0.9984
0.9983
0.9982
0.9981
0.9980
0.9979
0.9978
0.9977
TopTagXL AUC
10^7
10^8
10^9
Inference FLOPs, 50-particle jet (measured)
L-GATr-slim
LLoCa-Tr.
ParT
Transformer

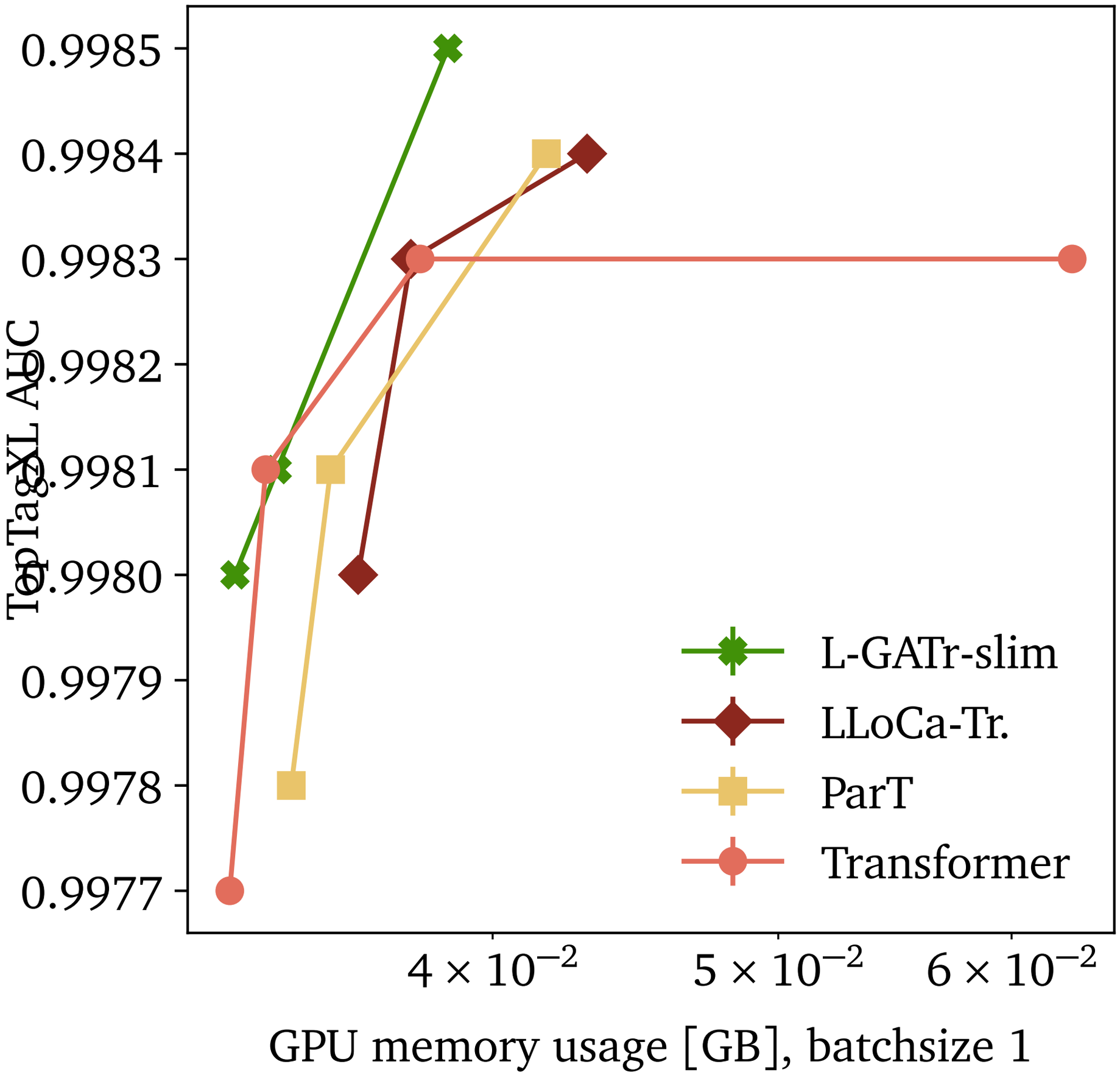

0.9985
0.9984
0.9983
0.9982
0.9981
0.9980
0.9979
0.9978
0.9977
TopTagXL AUC
10^7
10^8
Inference FLOPs, 50-particle jet (estimated)
L-GATr-slim
LLoCa-Tr.
ParT
Transformer

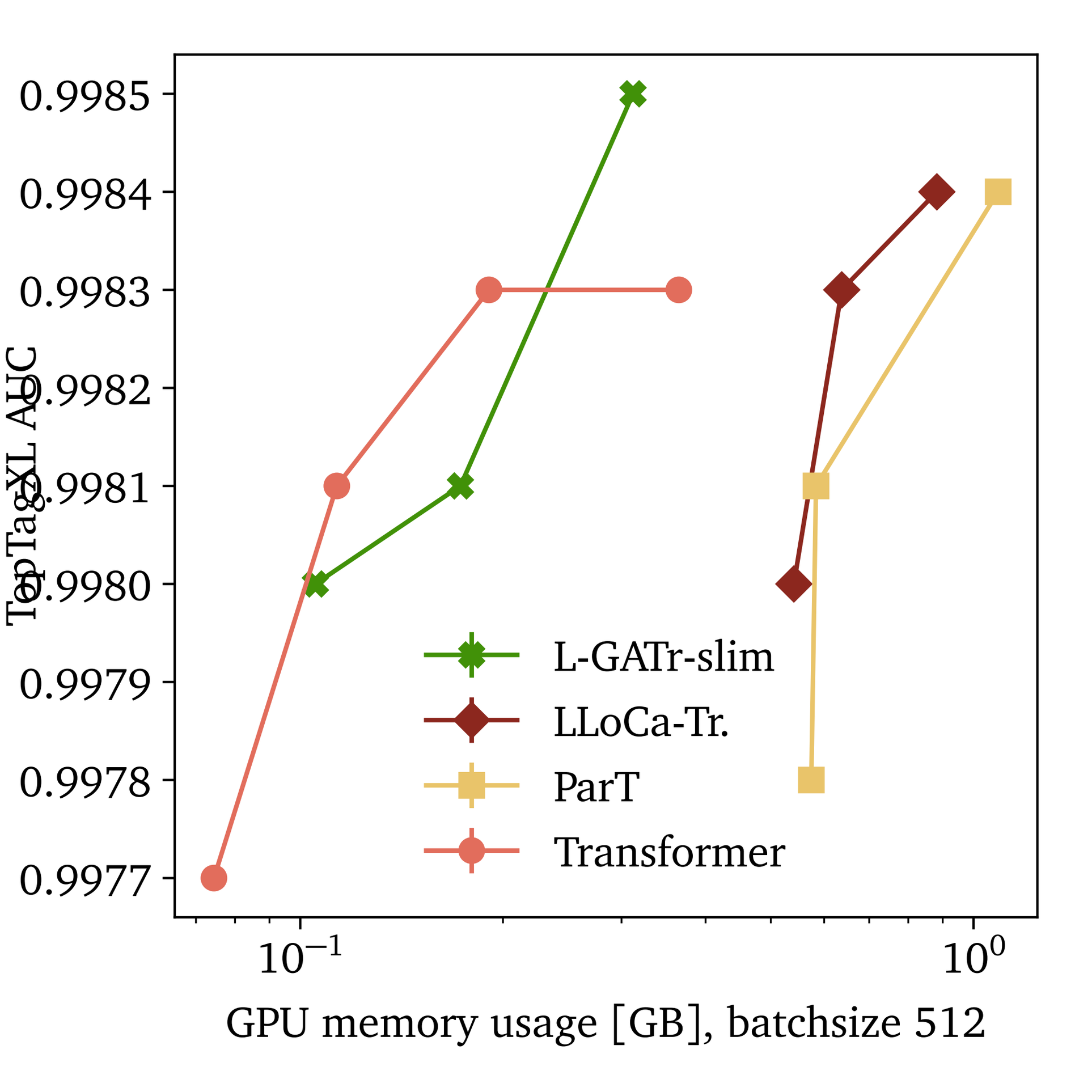

0.9985
0.9984
0.9983
0.9982
0.9981
0.9980
0.9979
0.9978
0.9977
TopTagXL AUC
10^7
10^8
Energy [pJ], 50-particle jet
L-GATr-slim
LLoCa-Tr.
ParT
Transformer

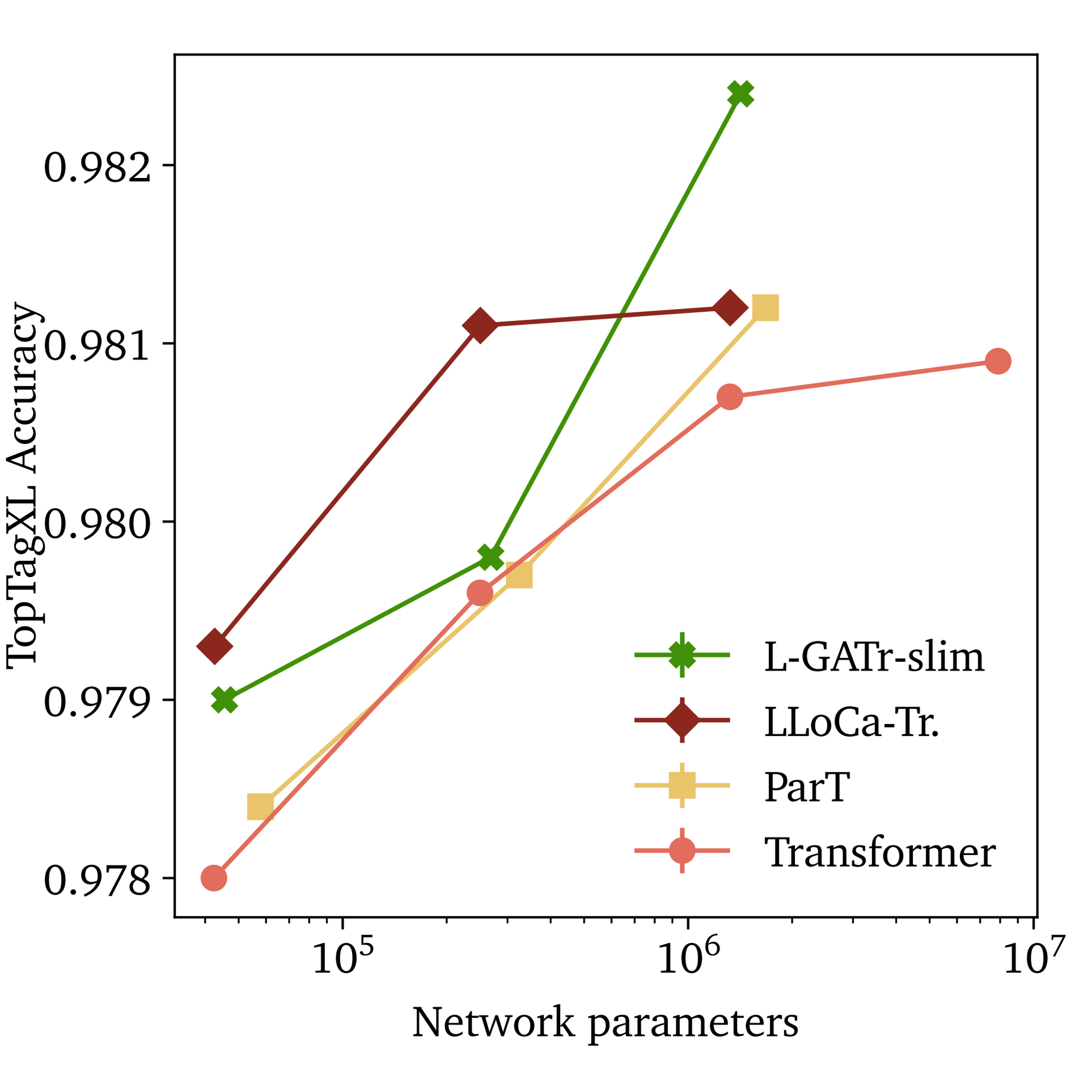

0.9985
0.9984
0.9983
0.9982
0.9981
0.9980
0.9979
0.9978
0.9977
TopTagXL AUC
3×10^0
4×10^0
6×10^0
10^1
CPU inference time [ms], 50-particle jet
L-GATr-slim
LLoCa-Tr.
ParT
Transformer

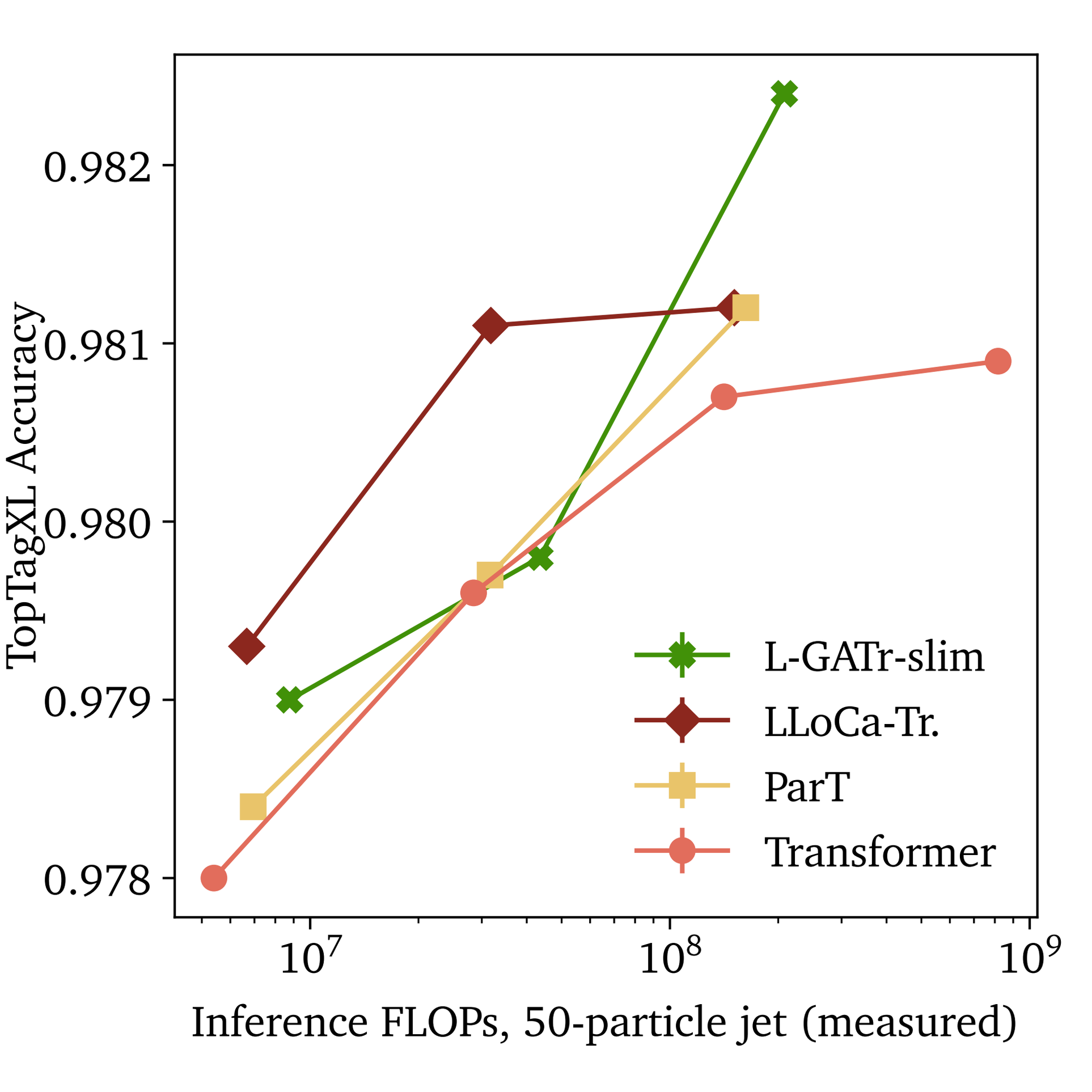

0.9985
0.9984
0.9983
0.9982
0.9981
0.9980
0.9979
0.9978
0.9977
TopTagXL AUC
5×10^0
6×10^0
7×10^0
8×10^0
9×10^0
10^1
GPU inference time [ms], averaged, batchsize 1
L-GATr-slim
LLoCa-Tr.
ParT
Transformer

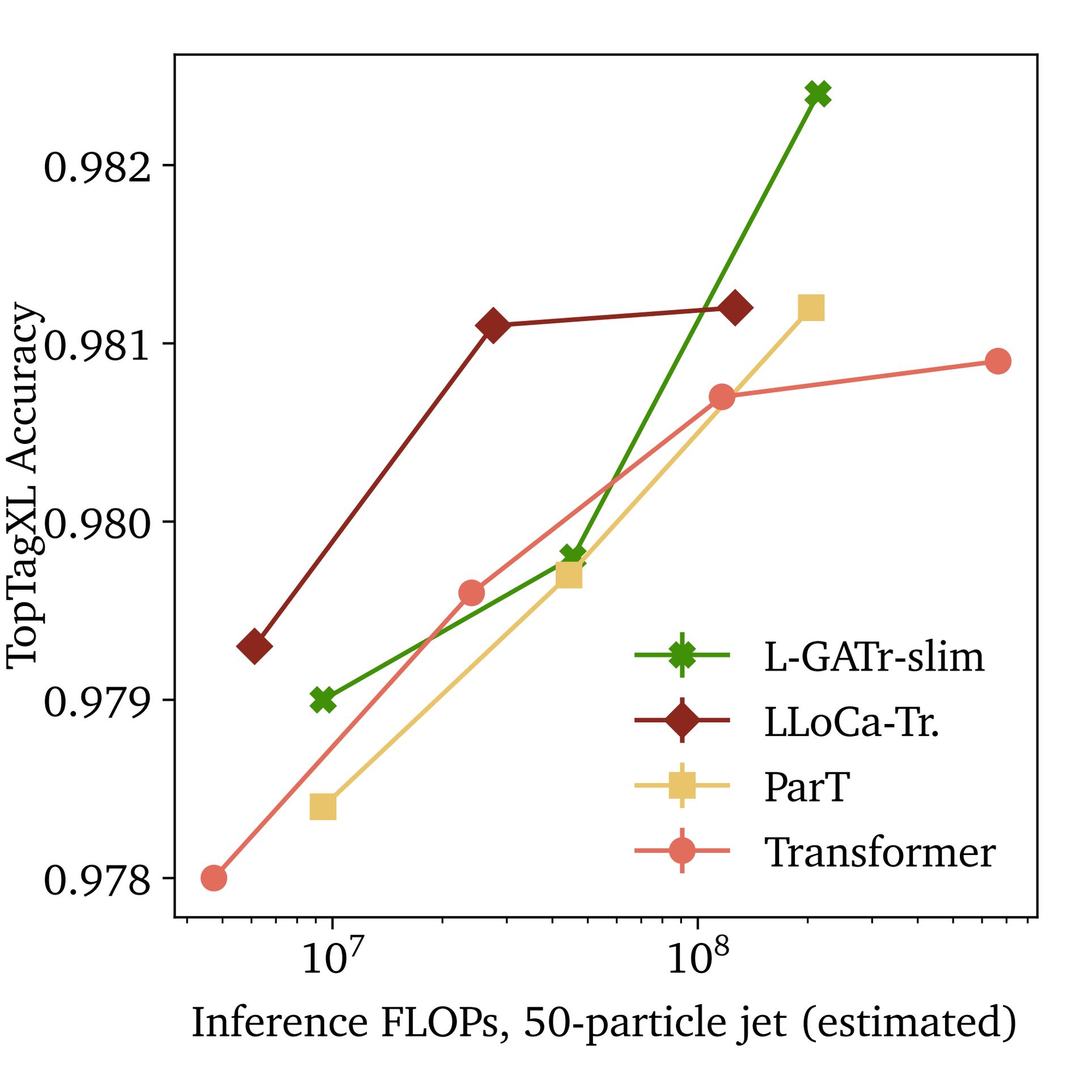

L-GATr-slim
LLoCa-Tr.
ParT
Transformer
0.9985
0.9984
0.9983
0.9982
0.9981
0.9980
0.9979
0.9978
0.9977
TopTagXL AUC
6×10^0
10^1
2×10^1
3×10^1
GPU inference time [ms], averaged, batchsize 512

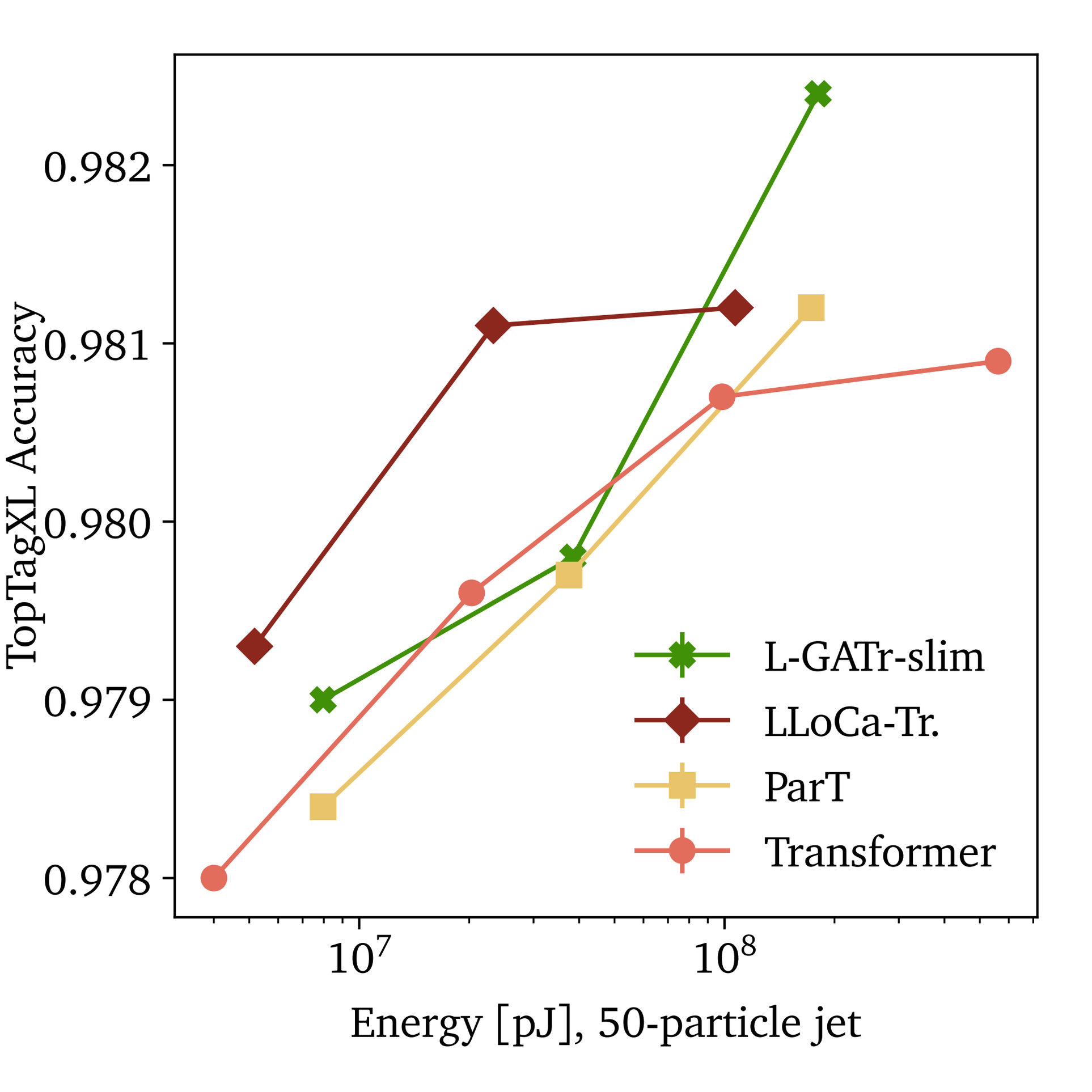

0.9985
0.9984
0.9983
0.9982
0.9981
0.9980
0.9979
0.9978
0.9977
TopTagXL AUC
4×10^{-2}
5×10^{-2}
6×10^{-2}
GPU memory usage [GB], batchsize 1
L-GATr-slim
LLoCa-Tr.
ParT
Transformer

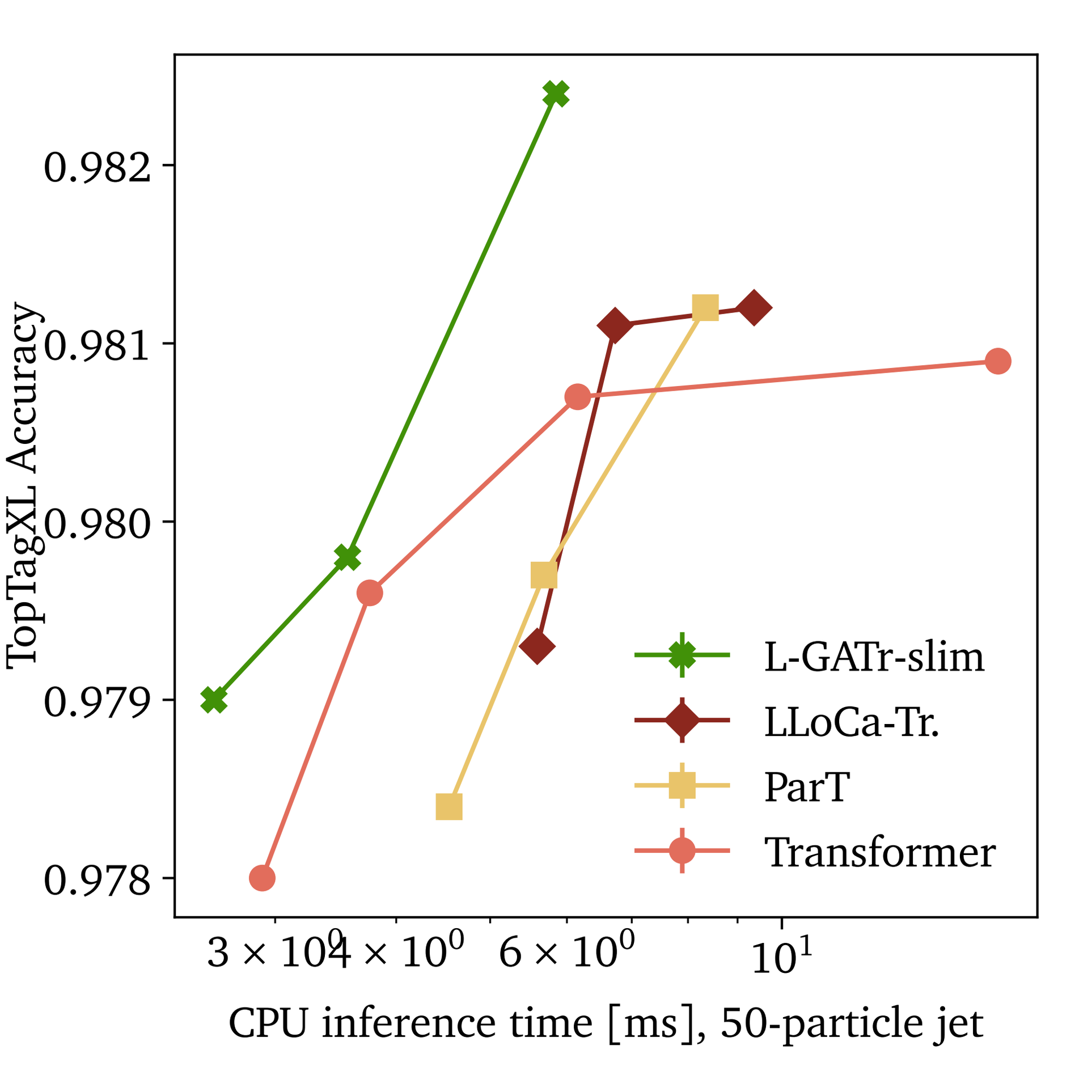

0.9985
0.9984
0.9983
0.9982
0.9981
0.9980
0.9979
0.9978
0.9977
TopTagXL AUC
10^{-1}
10^{0}
GPU memory usage [GB], batchsize 512
L-GATr-slim
LLoCa-Tr.
ParT
Transformer

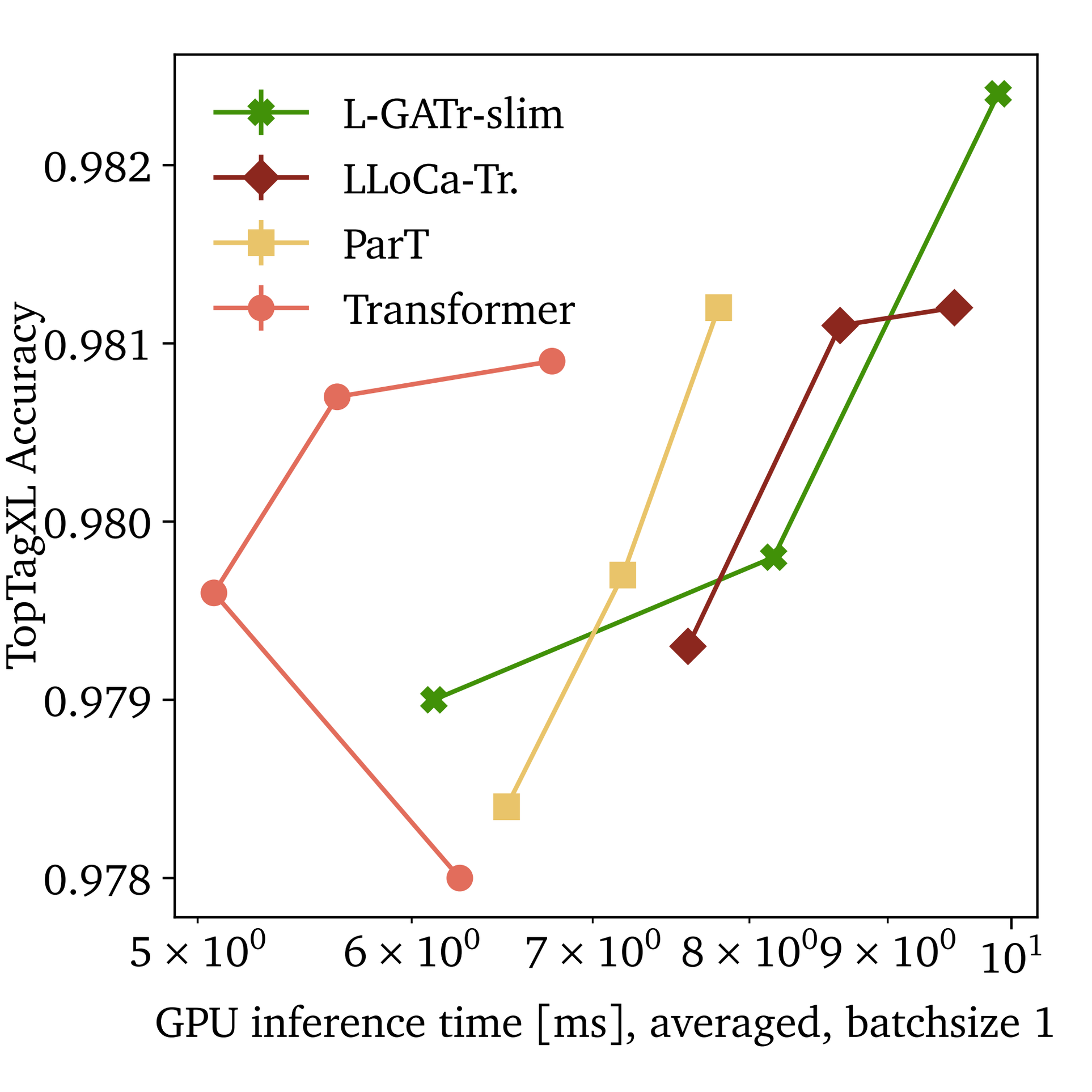

0.982
0.981
0.980
0.979
0.978
TopTagXL Accuracy
10^5
10^6
10^7
Network parameters
L-GATr-slim
LLoCa-Tr.
ParT
Transformer

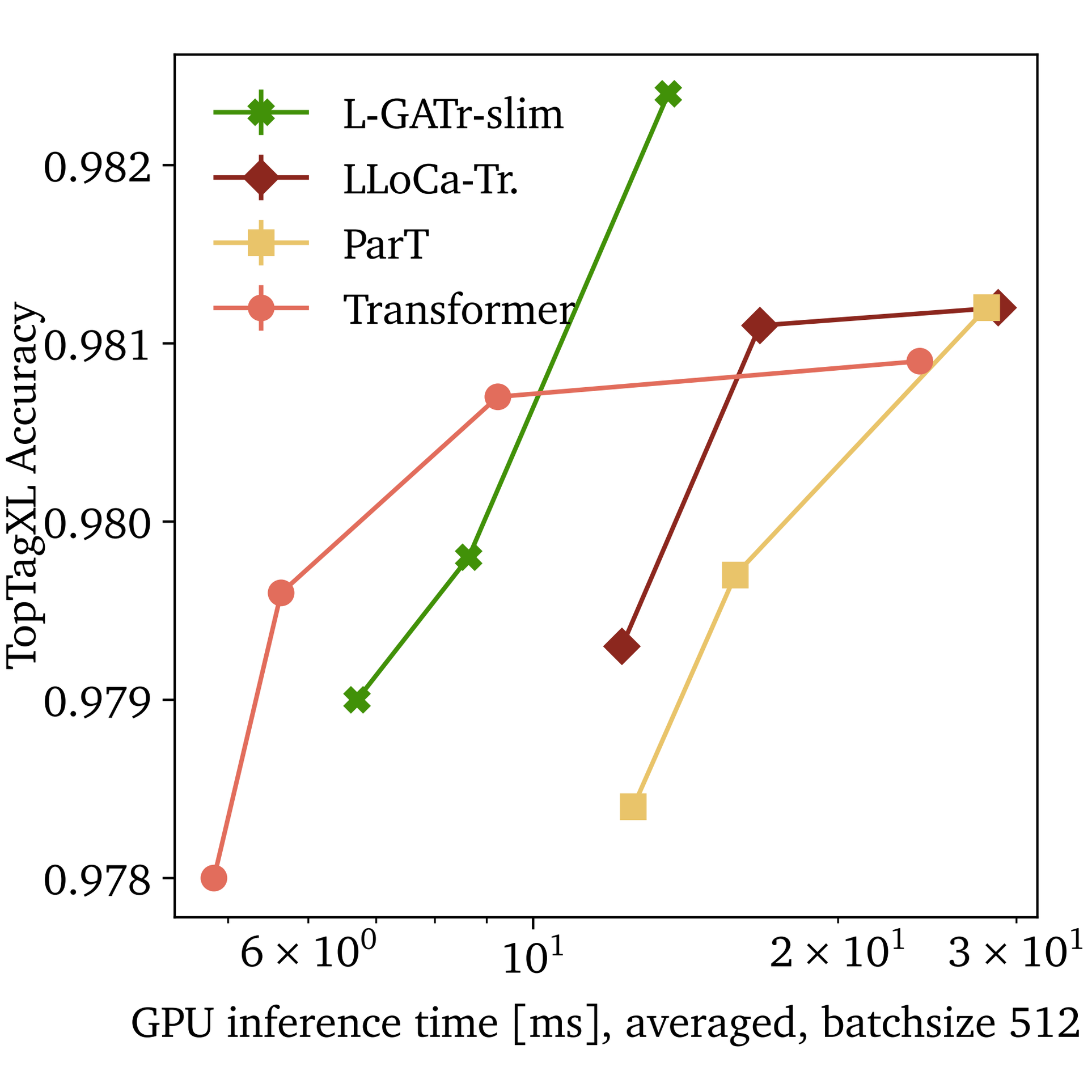

0.982
0.981
0.980
0.979
0.978
TopTagXL Accuracy
10^7
10^8
10^9
Inference FLOPs, 50-particle jet (measured)
L-GATr-slim
LLoCa-Tr.
ParT
Transformer

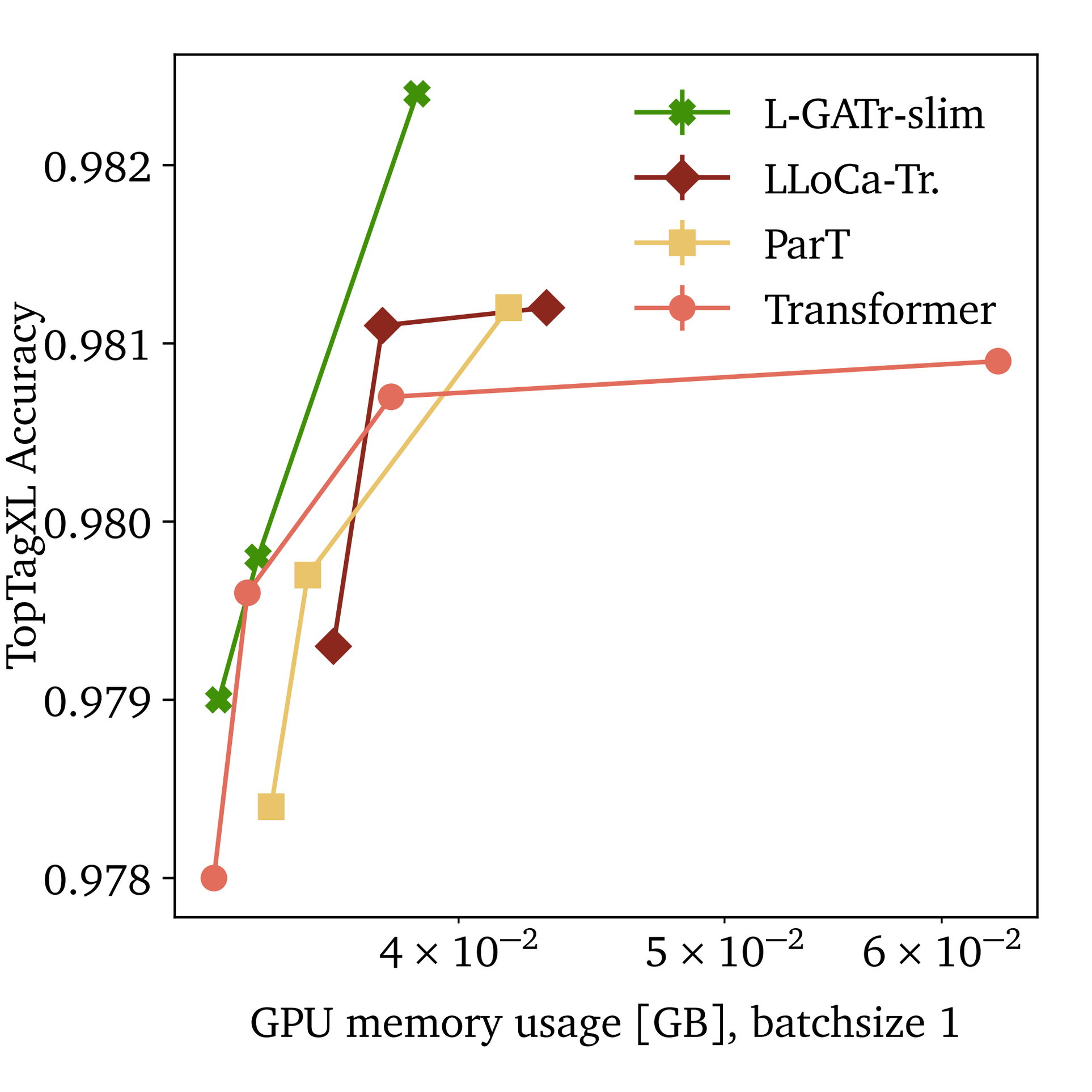

0.982
0.981
0.980
0.979
0.978
TopTagXL Accuracy
10^7
10^8
Inference FLOPs, 50-particle jet (estimated)
L-GATr-slim
LLoCa-Tr.
ParT
Transformer

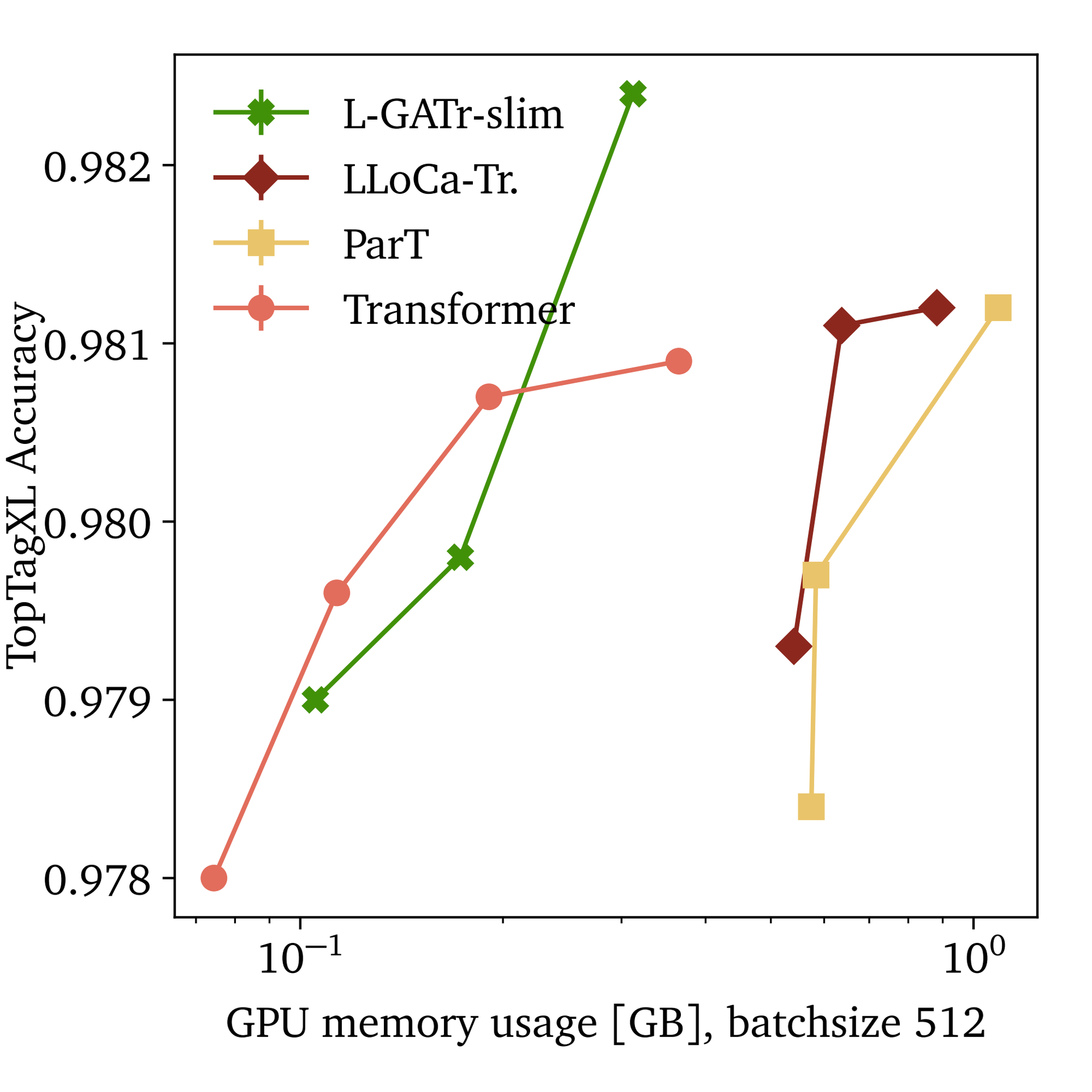

0.982
0.981
0.980
0.979
0.978
TopTagXL Accuracy
10^7
10^8
Energy [pJ], 50-particle jet
L-GATr-slim
LLoCa-Tr.
ParT
Transformer


0.982
0.981
0.980
0.979
0.978
TopTagXL Accuracy
3×10^0
4×10^0
6×10^0
10^1
CPU inference time [ms], 50-particle jet
L-GATr-slim
LLoCa-Tr.
ParT
Transformer


L-GATr-slim
LLoCa-Tr.
ParT
Transformer
0.982
0.981
0.980
0.979
0.978
TopTagXL Accuracy
5×10^0
6×10^0
7×10^0
8×10^0
9×10^0
10^1
GPU inference time [ms], averaged, batchsize 1


L-GATr-slim
LLoCa-Tr.
ParT
Transformer
0.982
0.981
0.980
0.979
0.978
TopTagXL Accuracy
6×10^0
10^1
2×10^1
3×10^1
GPU inference time [ms], averaged, batchsize 512


L-GATr-slim
LLoCa-Tr.
ParT
Transformer
TopTagXL Accuracy
0.982
0.981
0.980
0.979
0.978
4×10^{-2}
5×10^{-2}
6×10^{-2}
GPU memory usage [GB], batchsize 1


L-GATr-slim
LLoCa-Tr.
ParT
Transformer
0.982
0.981
0.980
0.979
0.978
TopTagXL Accuracy
10^{-1}
10^{0}
GPU memory usage [GB], batchsize 512